What is FAIR and AI-Ready for Datasets?

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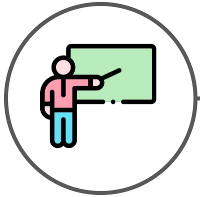
About This Presentation

20 min



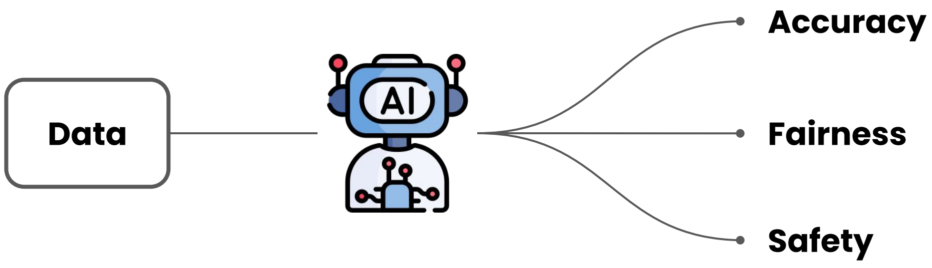
How to make research data FAIR and AI-ready

AI-Ready Data

AI-Ready Data

Data is the fuel for AI

Data enables the development of AI models and directly affects their characteristics



AI-Ready Data

Making research data AI-ready is important

If you share your research data
it is very likely to reused for AI development

<u>By making your research data-AI ready you can</u>:

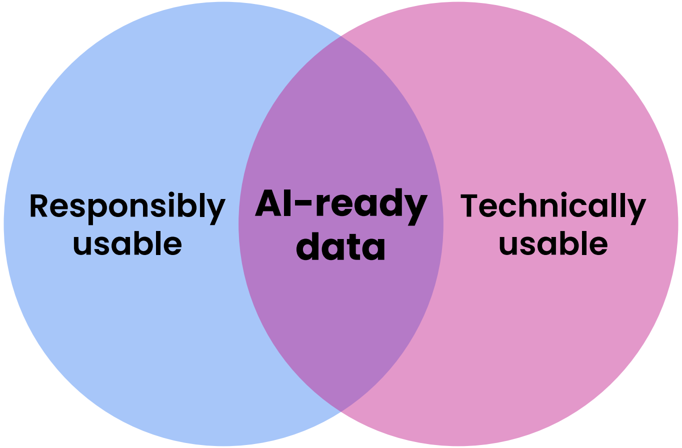
- Support and facilitate the development of AI models
- Ensure that your data is not (un)intentionally misused

AI-Ready Data

How to make data AI-ready?

- Protect participants privacy
- Ensure use align with consent
- And more

HIPAA, GDPR, CARE Principles



- Make data easy to find
- Use machine-friendly formats
- And more

FAIR Principles

FAIR Principles

FAIR Principles
Origin

How to make all research outcomes, including data, optimally reusable by humans and machines?

Findable, Accessible, Interoperable, and Reusable (FAIR) Principles (2016)

https://doi.org/10.1038/sdata.2016.18

FAIR Principles

15 principles to optimize data reuse for humans and machines

To be Findable:
F1. (meta)data are assigned a globally unique and persistent identifier
F2. data are described with rich metadata (defined by R1 below)
F3. metadata clearly and explicitly include the identifier of the data it describes
F4. (meta)data are registered or indexed in a searchable resource

To be Accessible:
A1. (meta)data are retrievable by their identifier using a standardized communications protocol
A1.1 the protocol is open, free, and universally implementable
A1.2 the protocol allows for an authentication and authorization procedure, where necessary
A2. metadata are accessible, even when the data are no longer available

To be Interoperable:
I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
I2. (meta)data use vocabularies that follow FAIR principles
I3. (meta)data include qualified references to other (meta)data

To be Reusable:
R1. meta(data) are richly described with a plurality of accurate and relevant attributes
R1.1. (meta)data are released with a clear and accessible data usage license
R1.2. (meta)data are associated with detailed provenance
R1.3. (meta)data meet domain-relevant community standards

https://doi.org/10.1038/sdata.2016.18

FAIR Principles
Interpretation

Findable → **Data is easy to find**
Data has a unique identifier (e.g. a DOI)
Metadata are provided and indexed in search engines

Accessible → **Data access process is clear**
Data is accessible using a standard protocol (e.g., HTTP)
Access requirements are clearly stated

Interoperable → **Data works well with related data**
Data and metadata are in standard, machine-friendly format
Data is connected to related resources

Reusable → **Data is easily usable**
Data has a clear usage license
Data is well documented

Very similar to what we do when sharing manuscripts!

FAIR Principles
What they are not

FAIR ≠ Open

FAIR ≠ Ethical or fair

FAIR Principles
Adoption

Leaders at the 2016 G20 meeting released a join press release expressing their intention to support implementation of FAIR principles in publicly funded research

“Turning FAIR into Reality” report (2018)

New data sharing policy (January 2023) requires all proposals to include a Data Management and Sharing Plan (DMSP) that describes how data will be made FAIR

FAIR Principles

How to practically make data FAIR

1. Structure data according to a consistent folder structure
2. Use standard, machine-friendly file formats
3. Include metadata files using standard format and vocabularies
4. Provide a data reuse license
5. Share/register dataset on a suitable data repository or registry

Steps 1–4 → Interoperable & Reusable

Step 5 → Findable & Accessible

The AI-READI Dataset

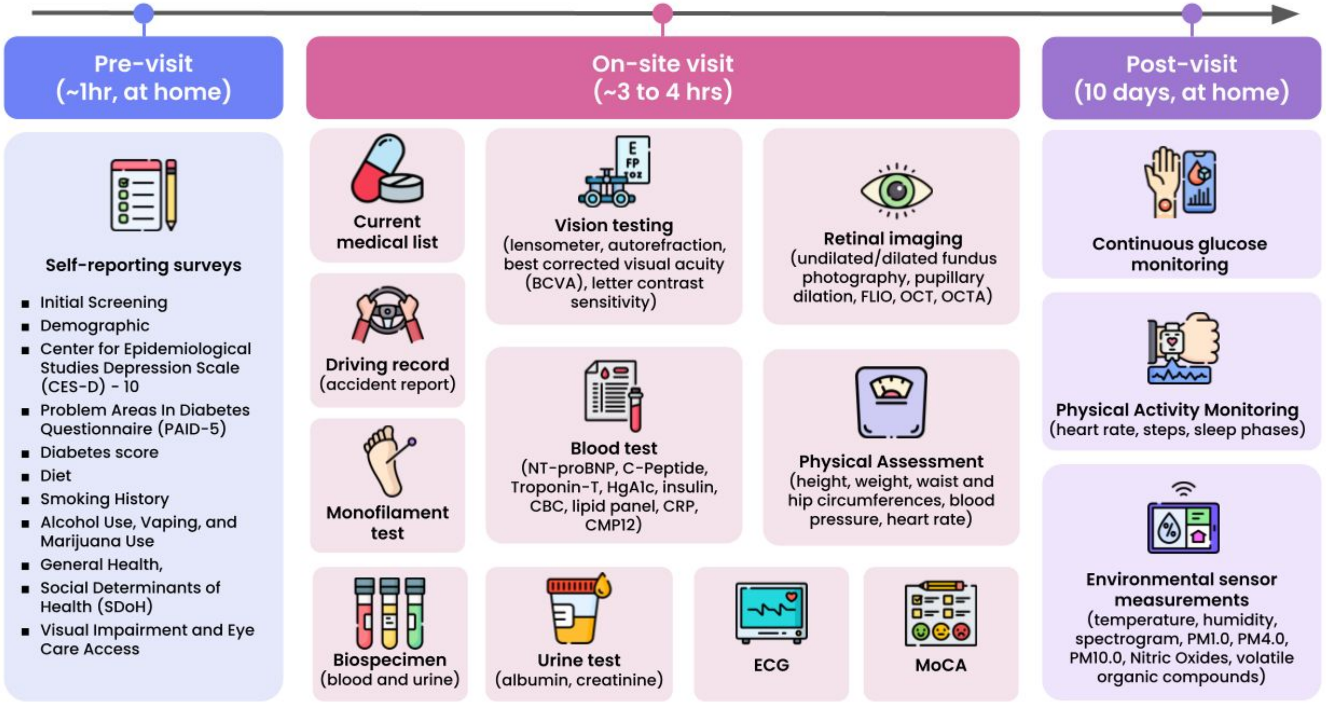
The AI-READI Dataset
About AI-READI

AI-READI is one of the Data Generation Project (DGP) of the NIH Bridge2AI Common Fund program

Goal: collect a multimodal dataset for studying Type 2 Diabetes and make it AI-ready

https://aireadi.org

The AI-READI Dataset
Data collection protocol



FLIO = Fluorescence Lifetime Imaging, OCT = Optical Coherence Tomography, OCTA = Optical Coherence Tomography Angiography,
ECG = Electrocardiogram, MoCA = Montreal Cognitive Assessment, PM1.0, 4.0, and 10.0 = Particulate matter less than 1, 4, and 10 microns, respectively

The AI-READI Dataset
Version 2 (November 2024)

Data from 1067 participants

165,000+ files

2 TB of data total

Target by the end of the project: 4,000 participants

https://fairhub.io/datasets/2

The AI-READI Dataset

FAIR: 1. Consistent folder structure

Clinical Dataset Structure (CDS)

cds-specification.readthedocs.io

- cardiac_ecg
- clinical_data
- environment
- retinal_flio
- retinal_oct
- retinal_octa
- retinal_photography
- wearable_activity_monitor
- wearable_blood_glucose

One folder per datatype

- CHANGELOG.md
- LICENSE.txt
- README.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- participants.json
- participants.tsv
- study_description.json

Metadata files

The AI-READI Dataset
FAIR: 2. Standard data format

Data category	Data format followed
Clinical data	Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)
Retinal imaging data	Digital Imaging and Communications in Medicine (DICOM)
ECG data (standard 12-lead)	WaveForm DataBase (WFDB)
Wearables data (physical activity monitoring, continuous glucose monitoring data)	Open mHealth
Environmental sensor data	Earth Science Data Systems (ESDS) format

The AI-READI Dataset

FAIR: 3. Extensive metadata

- retinal_photography
- wearable_activity_monitor
- CHANGELOG.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- LICENSE.txt
- participants.json
- participants.tsv
- README.md
- study_description.json

Broad metadata: study background, dataset provenance, dataset structure, license terms, participants information

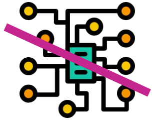
Provided in both human and machine-friendly formats

https://cds-specification.readthedocs.io

The AI-READI Dataset

FAIR: 4. Shared under a new usage license

Prohibits re-identification of study participants or harm



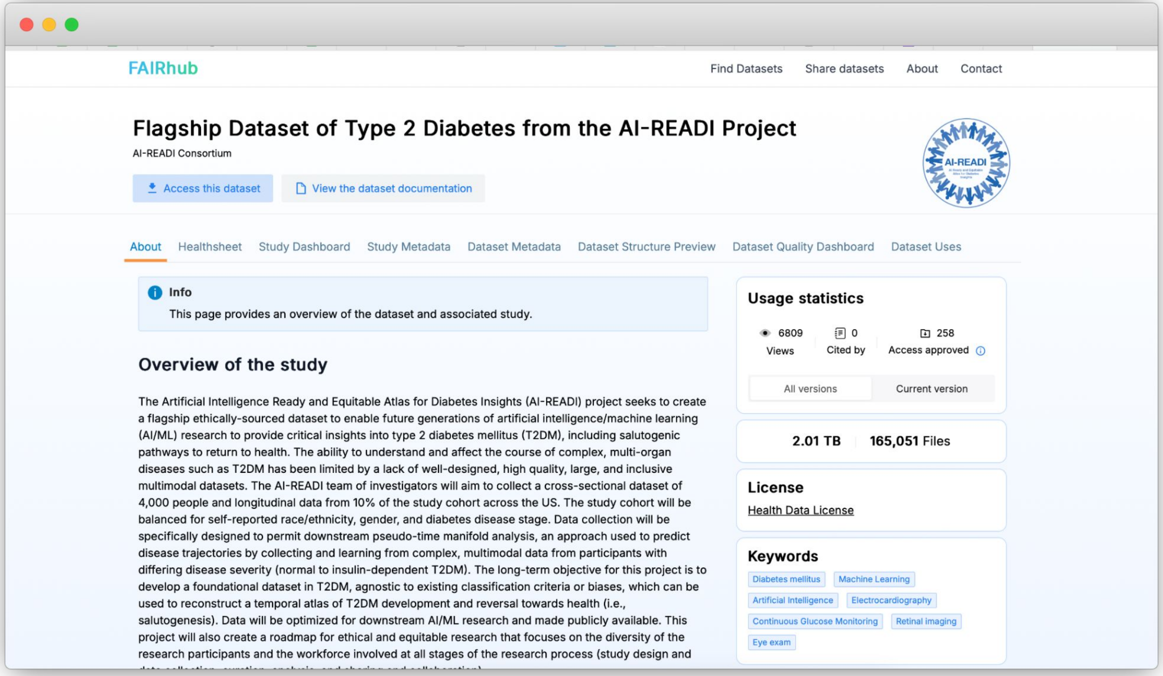
Prohibits distribution of the data and of models that may “copy” or “memorize” the data

Allows commercial use

https://doi.org/10.5281/zenodo.10642459

The AI-READI Dataset

FAIR: 5. Shared on FAIRhub



DOI

Accessible metadata

Indexed metadata

Clear access process

https://doi.org/10.60775/fairhub.2

Closing Remarks

Closing Remarks

Making data FAIR is not always easy…

› 1,000 standards

› 1,600 databases

› 100 policies

(“Bio” related results on FAIRsharing.org)

Closing Remarks

... but you can get started today with simple steps!

Kids		Share/register dataset on a suitable repository or registry	NIH list of data repositories
Adults		Provide as much metadata as possible	For example: authors, funding source, keywords, devices used, etc.
Legends		Structure (meta)data following standards	fairsharing.org

Closing Remarks

FAIR is good, AI-ready is better!

New Results

Follow this preprint

AI-readiness for Biomedical Data: Bridge2AI Recommendations

Timothy Clark, Harry Caufield, Jillian A. Parker, Sadnan Al Manir, Edilberto Amorim, James Eddy, Nayoon Gim, Brian Gow, Wesley Goar, Melissa Haendel, Jan N. Hansen, Nomi Harris, Henning Hermjakob, Marcin Joachimiak, Gianna Jordan, In-Hee Lee, Shannon K. McWeeney, Camille Nebeker, Milen Nikolov, Jamie Shaffer, Nathan Sheffield, Gloria Sheynkman, James Stevenson, Jake Y. Chen, Chris Mungall, Alex Wagner, Sek Won Kong, Satrajit S. Ghosh, Bhavesh Patel, Andrew Williams, Monica C. Munoz-Torres the Bridge to Artificial Intelligence (Bridge2AI) Consortium members

doi: https://doi.org/10.1101/2024.10.23.619844

This article is a preprint and has not been certified by peer review [what does this mean?].

Abstract | **Full Text** | Info/History | Metrics | Preview PDF

Abstract

Biomedical research and clinical practice are in the midst of a transition toward significantly increased use of artificial intelligence (AI) and machine learning (ML) methods. These advances promise to enable qualitatively deeper insight into complex challenges formerly beyond the reach of analytic methods and human intuition while placing increased demands on ethical and explainable artificial intelligence (XAI), given the opaque nature of many deep learning methods.

https://doi.org/10.1101/2024.10.23.619844

Thank You!

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fairdataihub.org

Find these slides and all resources here

bit.ly/FAIR-AI-ready